

Cross-language numerical comparison

Repository source: `expected/CROSS_LANGUAGE_COMPARISON.md`

The Python stress test was executed in the assistant's container. The matching base-R stress test was executed by Bruce Swihart in his local R environment. The two programs are independent implementations and use language-specific random-number generators, so they do not test the same random tensors even though both runs are deterministic within their own language.

Check	Python	R	Interpretation
Maximum pointwise determinant discrepancy	1.819e-12	2.046e-12	Direct projected-matrix determinant and compact tensor-contraction formula agree to floating-point precision.
Maximum relative error in <code>integral A_Y ^2</code>	2.829e-15	2.038e-15	Numerical quadrature agrees with <code>(176*pi/7)*tau</code> .
Maximum relative error in <code>integral (4 det A_Y)^2</code>	3.427e-15	5.107e-15	Numerical quadrature agrees with the proposed quartic invariant identity.
Candidate volume coefficient	0.383602704709067716	0.383602704709068	Agreement to all 15 digits printed by R.

All programmed numerical-stress acceptance thresholds passed in both languages. The new finite pairing-count table has passed the Python exact enumerator; the matching base-R enumerator is present but still awaits Bruce Swihart's local run.

This is strong cross-language numerical evidence for the tensor conventions and constants. It is not a proof of the full lower bound. In particular, the general symbolic integration is currently implemented only in Python/SymPy, and the full argument still requires a publication-ready LaTeX write-up and outside review. The bridge and degree-3 tensor derivations now exist as detailed Markdown proof drafts.